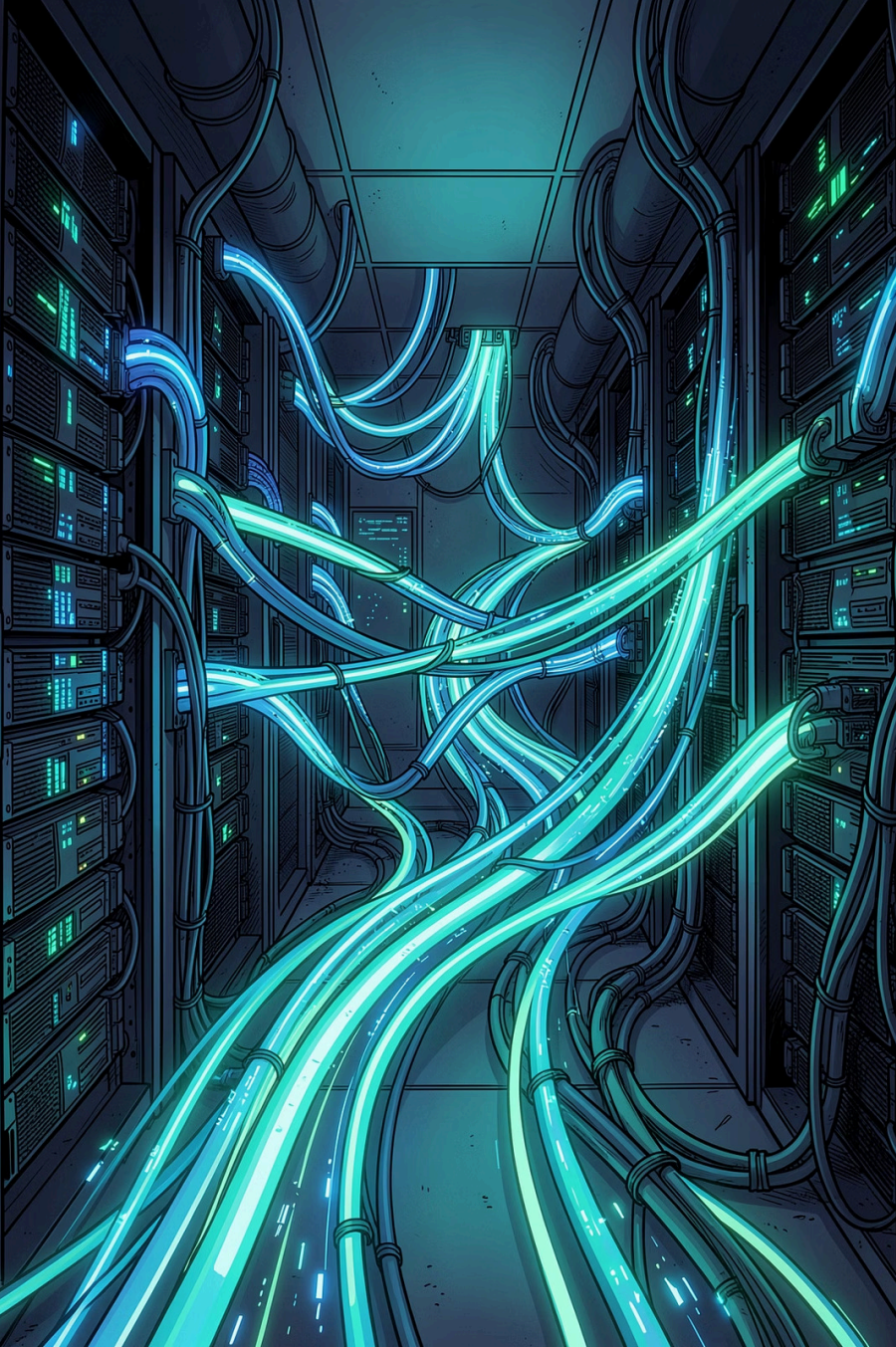




Intelligent Customer Support Analytics: A Hybrid Real-Time & Batch Data Pipeline

"A Scalable Infrastructure for Real-Time Sentiment Analysis and Agent Performance Tracking"

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The Challenge: Unstructured Support Data at Scale

The Problem

Technical support teams process thousands of unstructured tickets daily, causing critical delays in detecting systemic issues, tracking sentiment trends, and measuring agent KPIs objectively. Without automated pipelines, business intelligence remains reactive rather than proactive.

Target Users

- **Customer Support Managers** — real-time team oversight
- **Operations Executives** — strategic KPI visibility
- **Support Agents** — performance benchmarking

Value Proposition

Maximize support efficiency, prevent data drops, and deliver immediate automated business intelligence through real-time analytics.

High-Level Architecture: Source to Serving Layer



This architecture illustrates the complete data flow from multiple support data sources through real-time ingestion, distributed stream processing, analytical storage, and visualization layers. It demonstrates how raw ticket events are transformed into actionable insights using modern data engineering technologies.

Heterogeneous Data Ingestion Architecture

The 3 Core Data Sources



Real-Time Ticket Simulator (Python Ingestion)

Type: Real-Time Streaming Data

A custom Python script designed to simulate continuous, high-throughput customer support ticket events, streaming raw data directly into the system.



Customer Profile Lifecycle (PostgreSQL & Debezium CDC)

Type: Change Data Capture (CDC)

Real-time event capture using Debezium sitting on top of a transactional PostgreSQL database. It instantly tracks and streams background modifications, updates, and profile changes of the customers generated in Source 1.

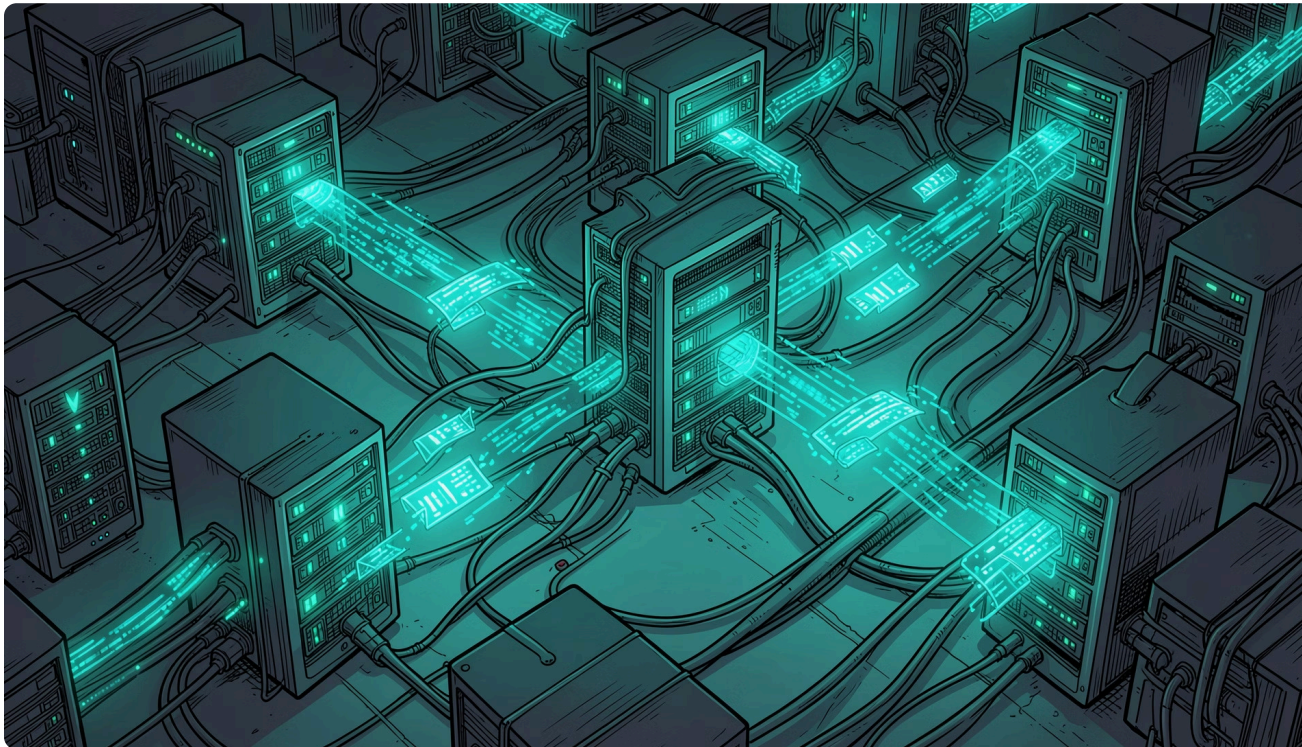


Operational ERP Ingestion (Frappe HelpDesk)

Type: Scheduled Batch Pipeline

An enterprise-level HelpDesk application powered by the Frappe framework. This represents the batch infrastructure, periodically synchronizing heavy historical support data, logs, and daily operational reports.

Data Ingestion & Streaming Infrastructure



Debezium CDC

Change Data Capture on transactional PostgreSQL databases captures row-level events instantly — without imposing load on production systems. Heterogeneous data formats from multiple sources are normalized at ingestion.

Kafka HA Cluster

- **3-broker** Highly Available cluster configuration
- **3 partitions** per topic for parallel throughput
- **Replication factor of 3** guaranteeing zero data loss under broker failure
- Advanced partitioning strategy ensures ordered, fault-tolerant event delivery

Distributed Stream Processing: Apache Spark

Spark Structured Streaming executes low-latency, real-time ETL on every incoming ticket event, enriching raw data with ML-derived features before persisting to storage.

1

NLP & Sentiment Analysis

Parses unstructured ticket content to extract real-time emotional classification: **Positive / Neutral / Negative**.

2

Advanced Feature Engineering

Extracts custom features, analyzes real-time sentiment, and dynamically calculates metrics like `current_age_minutes` to provide actionable insights that empower predictive decision-making and proactive ticket management.

3

Watermarking

Handles out-of-order event arrival by defining acceptable latency windows for late-arriving data.

4

Checkpointing

Ensures zero data loss with stateful recovery — Spark resumes from the last committed offset after any failure.

Analytical Storage Optimization: ClickHouse

Columnar Warehouse Engine

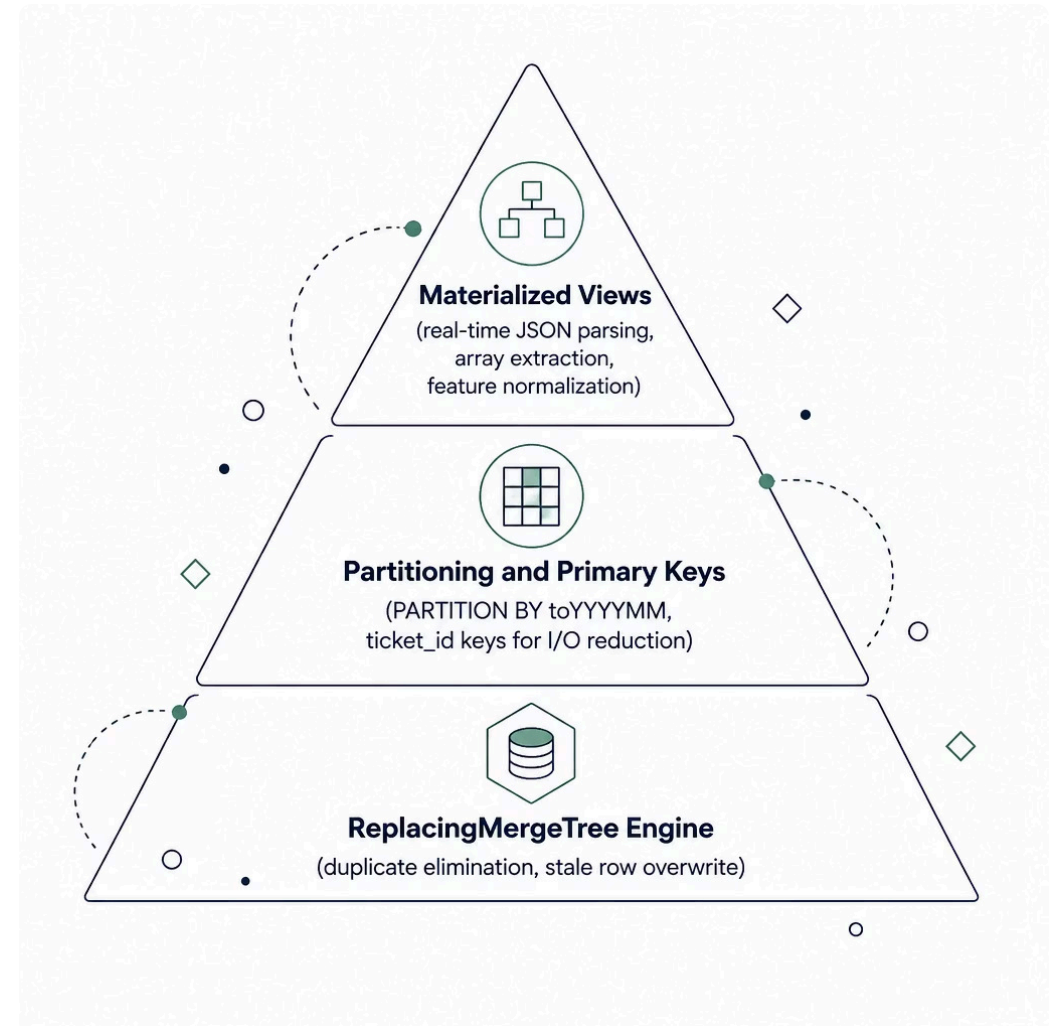
ClickHouse serves as the ultra-fast analytical backend, optimized for high-cardinality ticket data and sub-second OLAP queries across millions of rows.

Schema & Partitioning Strategy

- **ReplacingMergeTree** engine eliminates duplicates and overwrites stale rows by latest processing timestamp
- **PARTITION BY toYYYYMM** optimizes time-range query pruning
- Primary keys on `ticket_id` minimize disk I/O

Materialized Views

Pre-compiled views perform on-the-fly JSON parsing, array extraction, and feature normalization at ingestion — eliminating runtime extraction overhead.



Batch Pipeline & Hybrid Architecture: Frappe Integration

The system operates under a **dual-paradigm model** — combining real-time streaming with a scheduled batch pipeline to satisfy both operational immediacy and historical depth.

Real-Time Stream

Continuous ingestion and transformation of live ticket events via Kafka → Spark → ClickHouse. Supports sub-second KPI updates and live sentiment dashboards.

Batch Synchronization

Daily scheduled pipeline synchronizes ERP/CRM support metrics from **Frappe**, processes heavy historical datasets, and generates long-term agent KPI reports without impacting streaming throughput.

Business Intelligence: Grafana Analytics Dashboard



Agents Analysis

- **Agent Performance Leaderboard:** Tracks and ranks the most productive agents based on the total volume of successfully resolved and closed tickets.
- **Current Workload per Agent:** A live monitor displaying the real-time distribution of active tickets (Open and Replied) currently assigned to each agent to prevent burnout.
- **Agent Ticket Status Breakdown:** A stacked bar chart visualization showing the comprehensive lifecycle status of all tickets handled by each individual agent.



Live Sentiment Trends

- **Sentiment Distribution:** Real-time visualization of customer emotional categories across all incoming tickets (Positive / Neutral / Negative).
- **Negative Sentiment by Category:** A strategic panel pinpointing specific ticket types or product categories that trigger the highest customer dissatisfaction.
- **Sentiment Score Trend Over Time:** A time-series graph tracking the fluctuations of customer satisfaction levels over custom time windows to monitor support quality trends.



Resolution Time & Performance Metrics

- **Average Resolution Time:** A core high-level KPI calculating the precise average minutes taken to fully resolve and close support tickets.
- **Average Age of Open Tickets:** Tracks the live duration and accumulated delay of pending tickets that are still active in the system.
- **Average Resolution Time per Agent:** An advanced efficiency metric ranking agents by their speed and responsiveness in successfully closing tickets.
- **Ticket Status Breakdown:** A comprehensive operational view displaying the volume of tickets across all distinct states (Open, Replied, Resolved, Closed).
- **Top Critical & Oldest Tickets:** A proactive alerting panel highlighting high-priority, long-standing tickets that require immediate escalation.



Technical Challenges & Architectural Solutions

Challenge	Root Cause	Solution
Frappe & MariaDB Debezium CDC Integration	Schema incompatibilities and connection drops between Debezium connector and MariaDB container for Frappe.	Reconfigured the Debezium source connector properties to strictly support MariaDB binlog formats and aligned the user privileges.
Spark Streaming Windows & Hadoop Conflict	Running Apache Spark on Windows triggered fatal initialization errors due to missing native Unix-like binaries (winutils.exe) in the Hadoop eco-system embedded within spark-sql-kafka.	Manually integrated and configured the hadoop-3.3.5/bin path variables locally to bypass Windows runtime dependency failures.
Spark to ClickHouse Dependency Collision	Classpath dependency hell and conflicting library names when attempting to stream data from Spark to ClickHouse using standard JDBC jars.	Switched to the Shaded ClickHouse JDBC driver (clickhouse-jdbc-0.6.5-shaded.jar) to isolate internal dependencies and eliminate naming collisions.



CONCLUSION

Core Takeaways & Project Impact

Scalability Proven

The architecture demonstrates robust capacity to ingest, transform, and analyze **thousands of ticket metrics per second** — validating modern streaming data engineering patterns at enterprise scale.

Business Impact Delivered

Organizations gain a **data-driven framework** to optimize human resource assignment, reduce resolution latency, and proactively improve customer satisfaction through real-time sentiment and KPI visibility.

Hybrid Paradigm Validated

The coexistence of **real-time streaming** and **scheduled batch synchronization** proves that operational immediacy and historical analytical depth are not mutually exclusive — they are complementary.

- 📌 **Ready for Live Demonstration & Q&A.** All pipeline components — Kafka, Spark, ClickHouse, and Grafana — are deployed and operational for end-to-end walkthrough.